

MSC THESIS PROPOSAL

Exploring Amsterdam's Party Tourism: Employing the Theory of Planned Behaviour and Spatial Agent-Based Modelling to Understand its Current State and Envision its Future.



1 Photo credit: Andreas Kirchhoff

Femke van den Dries 1030949
Geo-Information Science
10/04/2024

Arend Ligtenberg (Supervisor)
Maryam Ghodsvali (Supervisor)



1. Introduction

While many destinations actively seek to attract tourists for their economic benefits, Amsterdam is trying to discourage them from coming (Gemeente Amsterdam, 2022c). The city has set a goal of limiting day-visitors and overnight stays to 20 million per year. However, for 2024 and 2025 this number is expected to be exceeded by millions, even under the most conservative estimates (Fedovora & Klingen, 2023). Overtourism has brought a disbalance to the numerous societal functions the city fulfils, the most important being a place to live. The mayor of Amsterdam went as far as stating that if nothing is done, no one will be living in the centre of Amsterdam anymore in one to two decades (Nikel, 2023). A significant proportion of the tourism problem is made up of the nuisance caused by the party tourists that the city attracts due to its' free image (NOS, 2022). Party tourists are characterized by shifting behavioural norms during the stay compared to those at home, for example in the area of alcohol, sex, or substance abuse (Gemeente Amsterdam, 2022d). Some negative consequences of this type of behaviour are that residents experience a lower level of safety, nuisance caused by drunks and waste in the streets (Sleutjes et al., 2022).

Numerous possible measures have been proposed to reduce the nuisance experienced by these party tourists. Some, like restricted opening times of clubs, have been accepted. Other high-impact measures are still under debate. These measures are abolishing window prostitution, charging an entrance fee to the red light district and installing an I-criterium, which implies that foreign visitors cannot buy weed legally from coffee shops anymore (Sleutjes & Bosveld, 2019). The debate is largely on the unknown effectivity and unforeseen consequences of the measures, like tourists switching to illegal alternatives (Brannan, 2022).

For the municipality to gain insight into the problem, tourism pressure and tourism related liveability have been mapped in detail. For tourism pressure, this has been done through indicators like number of coffeeshops per 1000 inhabitants. For tourism related liveability, inhabitants scored their experiences in surveys, e.g. how much nuisance by drunks they experienced (Sleutjes et al., 2020, 2022). Again through surveys, the municipality has tried to get an insight in the motivation of young party tourists and their possible reaction to certain measures like prohibiting window prostitution (Sleutjes & Bosveld, 2019). At first glance, these measures appear to address the root of the problem. However, it has not yet been envisioned how party tourism in Amsterdam would change because of these measures.

Envisioning these scenarios is not straightforward, as tourism is a complex system and therefore includes many interactions and behaves in a non-linear way (Boavida-Portugal et al., 2017; Farrell & Twining-Ward, 2004). Party tourism also has a strong social component, adding to the complexity. For example, it has been shown that risk-seeking behaviour is contagious amongst tourists (Zhang et al., 2024). If drugs are involved, tourists experience stronger social connections with their fellow travellers and have a more positive experience of their travels (Bingöl, 2022).

A practical solution to explore future scenario's despite this complexity is Agent-Based Modelling (ABM) (Boavida-Portugal et al., 2017). This method has been found to be very suitable to model tourism flows (Nicholls et al., 2016). It also enables the researcher to envision the future of various scenarios, making it a fit tool for policy research (Johnson & Sieber, 2009). In the past decade, ABM has therefore gained popularity as a tool to study transitions and management challenges (Hansen et al., 2019; Zhuo & Han, 2020).

2. Research needs

ABM is a bottom-up approach to modelling where agents and their behaviour forms the basis of the simulation. An agent can be for example a person or an institution and has a state which we want to represent. The state is what agents are, know or have. For example, satisfaction. The agents are present in an environment, which is everything that is not an agent but is relevant to their decision making (Shalizi, 2006). The actions of agents interact with the other agents and their environment. In turn this affects subsequent actions of the agent itself through altering its state. How the state is altered, and which states lead to which action, depends on rules. The environment, agents, states, rules and actions are modelled in an ABM to describe real-world processes (Macal, 2016). Contrary to top-down model methods, for example system dynamics models or discrete event simulation, ABM allows for the development of a model from concepts which capture the tourist behaviour which is at the base of the system (Johnson et al., 2017). It thus allows us to model the system based on previous research on tourist behaviour, before understanding the exact dynamics of party tourism in Amsterdam. Another advantage of ABM is that, due to this resemblance with the real world, the model is intuitive and can easily be explained to non-modelers, like policy makers. (Nicholls et al., 2016).

When applying ABM to Amsterdam, we can see the party tourists as agents. The environment of the agents, which is the city of Amsterdam, as well as the actions of other agents leads them to actions like smoking weed, sex tourism and consumption of alcohol through decision rules. This alters their state as each of these actions increases the satisfaction with their holiday. So far, there have not been developed any ABMs for a city with similar characteristics as Amsterdam, or for the tourist group that our research aims at: party tourists. Overall, the application of ABM in tourism research is rather recent, as it has only started to be picked up around 2010. The lack of tradition and low awareness of the method in tourism research have been barriers to widespread adoption (Boavida-Portugal et al., 2017).

There are some examples of existing applications in tourism. These roughly divide into two groups, The first group focuses on the processes regulating the tourist industry e.g. on destination choice (Alvarez & Brida, 2019) or competition between tour operators in Antarctica (Student et al., 2016). The second group implements a geographical component, showing behaviour of tourists spatially. Examples of this are the envisioning of climate change adaption strategies for the alpine tourism industry (Balbi et al., 2013) or how tourists move between islands in the Galapagos based on price changes (Pizzitutti et al., 2014). These ABMs take an economic or consumer-based perspective to tourism.

However, for the tourist group that this research aims to study, we assume the actions are not driven by economic behaviour. Instead, we see their behaviour is influenced by the behaviour of others (Zhang et al., 2024) and the tourists' perception of the city (Gemeente Amsterdam, 2022b). ABMs are shown to be useful for modelling behavioural changes through policy, as they can simulate the influence of a changing environments on agents' actions. These changes in actions added up, may create a new behavioural pattern in the group of agents (Kasargodu Anebagilu et al., 2021; Pakravan & MacCarty, 2020).

To model behavioural change, we first need to understand tourists' motivations for exhibiting certain behaviour. The Theory of Planned Behaviour (TPB) is proposed in literature as a suitable framework to describe motivations and predict subsequent behaviours of tourists (Duarte Alonso et al., 2015; Juvan & Omerzel, 2017; Seow et al., 2017). This theory splits out the motivations leading up to behaviour into 3 components: Attitude, Subjective Norm & Perceived Behavioural Control (Ajzen, 1991). Previous work has also shown the possibility to translate this theory for use in ABMs (Kasargodu Anebagilu et al., 2021; Pakravan & MacCarty, 2020). However, the factors that compose the TPB are specific to characteristics of agents and their circumstances.. The implementation in our model thus depends on the components

which make up the TPB for party tourists. This will be based on both more general work, describing the behaviour and motivations of party tourists (Bingöl, 2022; Zhang et al., 2024), as well as work looking into the specific motivations of tourists in Amsterdam (Gemeente Amsterdam, 2022d; Sleutjes & Bosveld, 2019).

In our approach we combine this previous work on spatial tourism flows and behavioural modelling. The aim is to envision how changes in the environment (policy changes) may lead to a new spatiotemporal pattern of party tourists throughout the city. The societal need is not only to understand the dynamics behind party tourism, but also to envision the effect of proposed policy measures with a large expected impact that are still under debate. As previously mentioned, doing so is not straightforward due to tourism being a complex system (Boavida-Portugal et al., 2017; Farrell & Twining-Ward, 2004). Policy measures as are currently proposed, will change the environment in which the agents (tourists) make their decisions for certain actions. This will affect the components of the TPB and consequently observed actions. As agents show actions in different locations or will cease to perform certain actions, this will in turn influence other agents due to the social component of the TPB. Testing the effect of policy measures will therefore build upon the patterns of behavioural decisions (actions) that are to be discovered through the model.



3. Overall research aim and research questions

The aim of this research is to understand through implementing the TPB into a spatial ABM how environmental and social factors in Amsterdam contribute to the behaviour of party tourists. In turn, these behaviours lead to a spatiotemporal pattern of party tourism in Amsterdam. With environmental factors we mean government policy, presence of hotels, clubs, and coffee shops and other factors which are found relevant in our research. For social factors we consider the presence of other tourists showcasing the same behaviour. In the ABM we model the factors in the tourist decision process determining the behaviour that can explain the current state of party tourism. What these factors are, is to be determined through literature research and empirical knowledge. This model then provides the basis to envision several proposed policy scenarios.

We see the following behaviours, in this thesis also referred to as actions, of party tourists we want to model:

- Smoke weed,
- Sex-tourism related activities,
- Use alcohol & illicit drugs.

This model will then provide a testing ground to assess different policy strategies to deter party tourism. Ultimately, the goal is to test several proposed measures, those are:

- Introduction of the I-criterium,
- Abolishing window prostitution in the Red-Light District and opening a new erotic centre,
- An entrance fee for the Red-Light District.

This aim leads us to the following main and sub research questions (SRQ):

‘How do components of the Theory of Planned Behaviour influence party tourists’ actions in an Agent-Based Model of Amsterdam, which subsequently contribute to the spatiotemporal pattern of party tourism in the city?’

SRQ1: What factors that influence party tourists’ actions in Amsterdam according to the TPB can be implemented into a spatial ABM?

SRQ2: To what extent do all factors that influence the tourists’ actions contribute to the currently observed spatiotemporal spread of party tourism when modelled in an ABM?

SRQ3: How do the proposed policy scenarios alter the model parameters and thereby the spatiotemporal spread of party tourism that is outputted by the ABM?

4. Methodology

4.1 Study area + data

The study area comprises the city of Amsterdam, capital of the Netherlands. This thesis will employ secondary data provided by the municipality of Amsterdam. One of the datasets they provide is on tourism pressure and tourism related liveability indicators for all neighbourhoods of the city for the years 2019 and 2021. For tourism pressure, this has been done through 7 indicators obtained from city registers. The most relevant for our research are number of beds in hotels, number of coffeeshops per 1000 inhabitants, number of sidewalks with little walking space due and number of foreign PIN transactions. For tourism related liveability, data was also obtained from city registers which store reports on e.g. safety and pollution. The most relevant indicators for our research are the nuisance by drunks and feeling of unsafety (Sleutjes et al., 2020, 2022). The source report divides the data into 4 quartiles, showing how the neighbourhoods perform on the mentioned indicators relative to each other. We will therefore also not interpret the output of our model as absolute numbers but will check whether we observe the same spatial distribution of party tourism as the data describes.

Although measurement year 2021 was affected by COVID, we see an increase in relative touristic pressure and a decrease in tourism related liveability for the busiest neighbourhoods compared to 2019.

Between 2019 and 2021 the city implemented shorter opening times and measures to discourage stag parties and organized pub crawls (Gemeente Amsterdam, 2022c).

Furthermore, we have survey data describing the motivations of tourists visiting the Wallen/Singelgebied to come to Amsterdam. As a part of this survey the participants were also asked how they would react to the introduction of an I-criterium, the abolishment of window prostitution and an entrance fee for the Red-Light District (Sleutjes & Bosveld, 2019). These factors describe the attitude of tourists to certain measures which forms a part of the TPB. Lastly, locations of landmarks, hotels, clubs, and other attractions found important to party tourism will be taken from Openstreetmap.

4.2 Model development

The design of the ABM will be set-up according to the ODD+D protocol by Müller et al. (2013). This methodological protocol is based on the Overview Design Details (ODD) protocol by Grimm et al. (2010) frequently used to describe ABM. By giving guidelines for a detailed and comprehensive description, the protocol promotes the reproducibility of the model. The ODD+D provides an additional framework to describe the decisions inherent to ABM's with a human component (Müller et al., 2013). It does so by giving guiding questions on structural elements of the model, e.g. the purpose of the model or how learning is implemented for the agents.

The model will be set-up in NetLogo, this is an open-source multi-agent modelling environment. The GIS extension of NetLogo will be employed to implement spatiality into the model. Spatial attributes that will be included in the model are those that in SRQ1 are found to be relevant to tourist behaviour.

In Figure 2 on the following page you find a schematic representation of the methodology for the model development and answering of the research questions.

Schematic representation Methodology

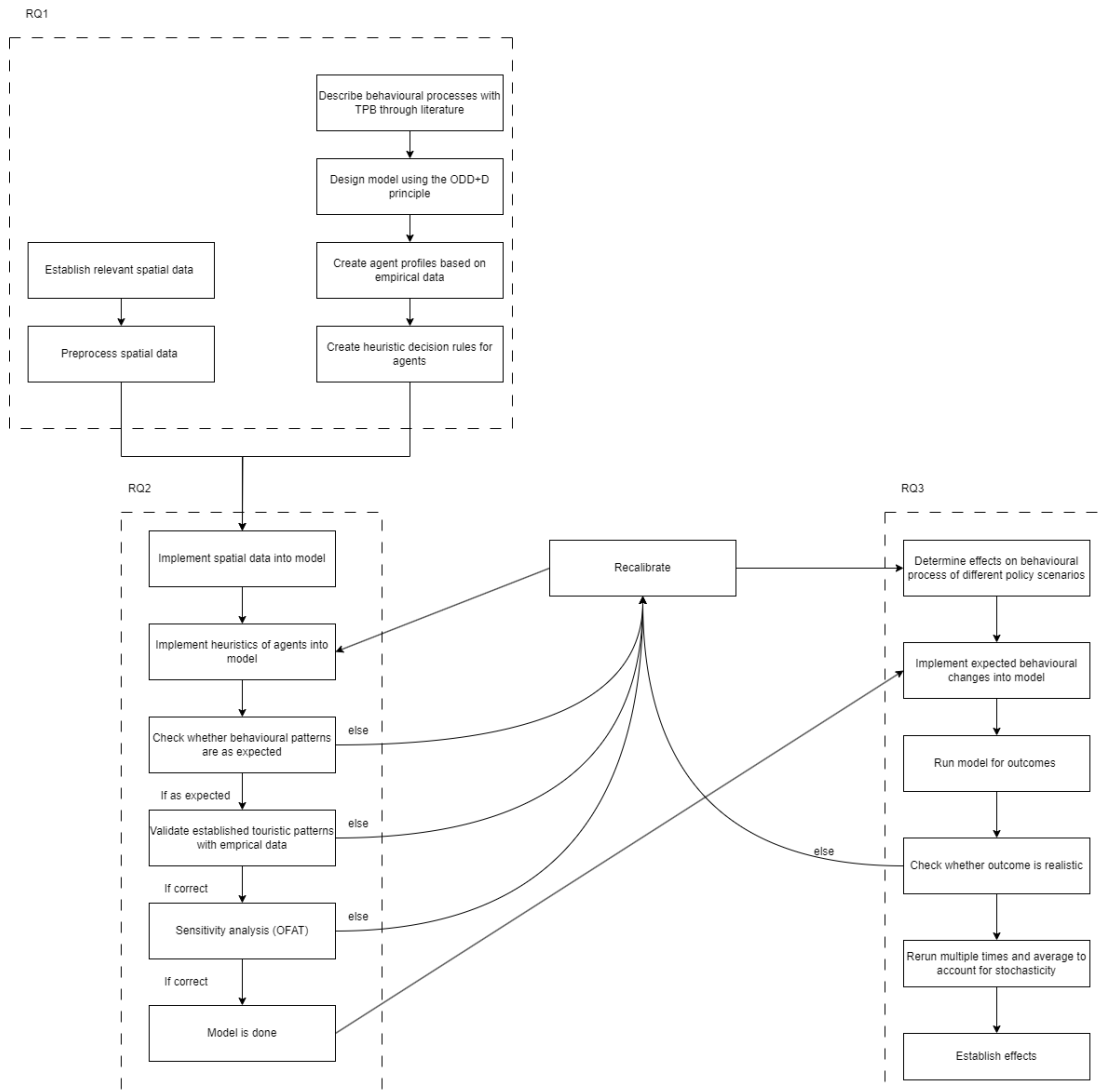


Figure 2 Schematic representation of methodology

4.2.1 Explanation of tourist behaviour through TPB (SRQ1)

SRQ1: What factors that influence party tourists' actions in Amsterdam according to the TPB can be implemented into a spatial ABM?

In this model we want to further understand the behaviour of party tourists in Amsterdam. To do so, we will introduce agents into the model that are all showcasing one type of behaviour. This simplification is justified because we are interested in seeing the pattern of behaviours around the city and not individual behaviours. One physical individual could in this methodology be represented by several different agents, composing its' tourist profile (e.g. a tourist who smokes weed and consumes alcohol & illicit drugs is represented by combining agents in group 1 & 3). This is possible as the ticks represent days, and in practice multiple activities could be undertaken for one day. Again, these actions/behaviours are:

1. Smoke weed,
2. Sex-tourism related activities,
3. Use alcohol & illicit drugs.

The behavioural rules of the agents will be based on a rule-based decision path, based on the components of the Theory of Planned Behaviour. In this theory, the behaviour of an agent is a function of the Attitude, Subjective Norms and Perceived Behavioural controls. Figure 3 describes the behavioural process which determines whether the agent participates in the activity it has been assigned to.

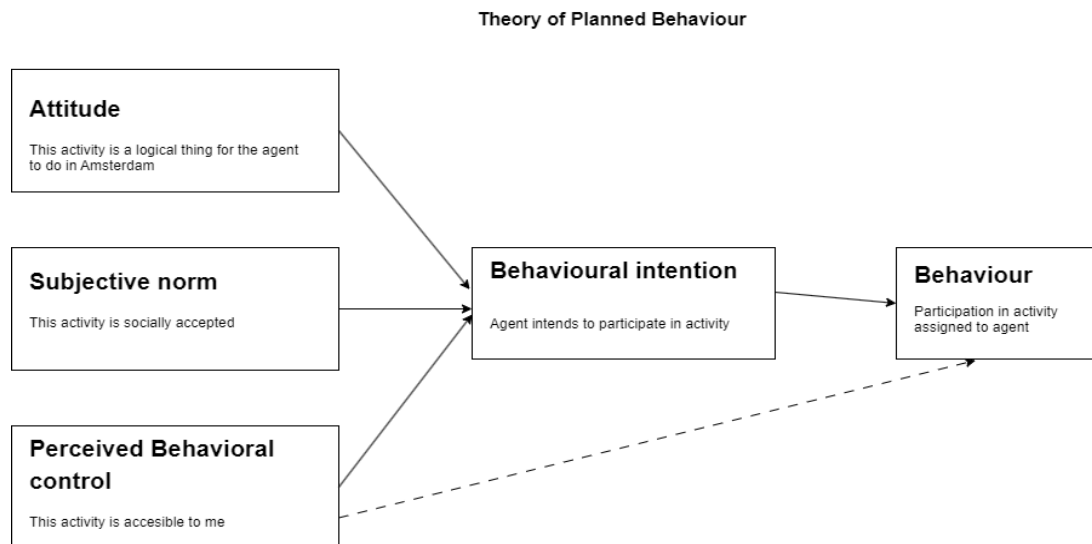


Figure 3 Decision scheme based on TPB for touristic activities in Amsterdam

The attitude, subjective norm and perceived behavioural control of agents will be represented by proxies. The following proxies are proposed for implementation into the model:

- **Attitude (AT)**: The results of the study by Sleutjes & Bosveld (2019) in which agents were queried on how important the presence of coffeeshops, bars and cafes and window prostitution were to come to the city. In the baseline situation, we consider the attitude of the agents to be enabling to the activity. In case of certain measures, we expect this factor to reduce the number of agents by the percentage of interviewees that indicated they would never visit the city anymore based on certain measures.

- **Subjective Norm (SN)**: The number of other agents in spatial proximity doing the same activity. It has been shown that risk-seeking behaviour is copied from other tourists (Zhang et al., 2024). We therefore assume that this subjective norm drives the growth of party tourism. For the baseline situation, we also assume the growth of party tourism will match the growth of 6,4% percent that is in line with the pre-covid growth of tourism (2016-2019) in Amsterdam (Gemeente Amsterdam, 2022a). Proximity has yet to be defined.

- **Perceived Behavioural Control (PC)** The number of elements in the environment of the model enabling the activity. This is the presence of coffeeshops, the presence of clubs and the presence of window prostitution. It is also affected by positive past experiences, so could implement the satisfaction score. PC is also the link between intention and actual behaviour, some agents may intend certain behaviour, but feel it is unsafe or impossible to do so.

To answer this research question, rule-based decision paths based on the components of the TPB will be formulated per group of agents. All these rules will be formulated to have a link with the environment, the state of the agent or the other agents (through proxies), so that they can be implemented into the ABM. This formulation of these paths is motivated by literature and data. The proxies proposed in the section above serve as a starting point. Calibration of the rule-based decisions is part of SRQ2.

4.2.2 Simulation of tourist behaviour in ABM (SRQ2)

SRQ2: To what extent do all factors that influence the tourists' actions contribute to the currently observed spatiotemporal spread of tourism when modelled in an ABM?

To answer this research question, the model will have to be set up as described in section 4.2. For the initialization agents will be placed in the model randomly. All agents start with a standard satisfaction score (state) which is positive, e.g. 8. In RQ1 we will have formulated rule-based decisions based on proxies. For example: For an agent of group 1 to perform its' behaviour (smoke weed) it will need to have a coffeeshop in proximity, it will need a few other agents doing it and it should not be prohibited by law to smoke weed.

For RQ2, these decision rules are to be implemented and calibrated in the model. For example: if a coffeeshop is present within 500 meters AND at least 10 other agents but less than 100 are present within 100 meters AND agent has a positive attitude >> agent of group 1 will perform activity, which leads to a +1 score in satisfaction. IF NOT: Agent will move for the following tick according to a random path, -1 score in satisfaction. If an agent reaches a negative satisfaction score, it will leave the simulation. When a certain number of agents is satisfied, an extra agent will be introduced. This example serves as an explanation and is not yet calibrated. As behaviour still has an unpredictable aspect, stochasticity will be introduced into the model. This is done by setting the parameters of the decision rules to a mean with a standard deviation. The parameters used for the decision rules of a certain agent in the model will then be a random draw from the normal distribution (Balbi et al., 2013).

Also, heterogeneity within the group needs to be ensured in the model. Not all agents belonging to one group will need the same conditions to show their behaviour as is also indicated in the survey results (Sleutjes & Bosveld, 2019). This will be done by creating several different types of agents with different decision rules within each group. It is also reasonable to assume that agents consider only a limited number of factors in practice, as they experience bounded rationality (Manson, 2006).

To calibrate the different components of the behavioural model (AT, SN, PC), the parameters will be adjusted to show the expected response whilst keeping constant all other variables. The outcome of the model can be compared to empirical data on tourism pressure (Sleutjes et al., 2020). For simplicity, we only look at the demand factor and assume the supply side only changes through policy changes, not through changes in demand. The calibration of the model is an iterative process, where it will be checked whether the behaviour is as could be expected from empirical knowledge after implementation of each agent (see Figure 3).

As a final step, the sensitivity of the model will be tested. Sensitivity analysis gives insight into the robustness of the model and how the parameter settings influence the outcome. This will be done through one-factor-at-a-time (OFAT) testing. This type of testing changes one parameter at a time, keeping all others constant. It is fit to show whether the outcome of the model depends on strong assumptions made for individual parameters (ten Broeke et al., 2016).

Testing policy measures (SRQ3)

SRQ3: How do the proposed policy scenarios alter the model parameters and thereby the spatiotemporal pattern of party tourism that is outputted by the ABM?

The simulation of tourist behaviour as performed for RQ2 will form the basis for testing the policy scenarios for RQ3. The policy scenarios will change the ABM environment. The policy scenarios will be

implemented as settings in the model, where the model can immediately be executed with certain policy scenarios. The effect on the behaviour of the agents will be because the outcome of the decision rules of the agents will partially depend on the environment, as well as on the other agents.

PCB is the component of the modelled behaviour that is directly affected by policy. For example, closing of coffeeshop will lead to a complete depletion of behavioural control for some agents of type 1 not leading to the activity (False response to condition coffeeshop is accessible within 500 meters) and ultimately leading to negative satisfaction, thus removal from the model. However, some agents of type 1 will still perform their activity, smoking weed, as some of the tourists they represent indicated they would still buy weed from drug dealers to bypass the measure (Sleutjes & Bosveld, 2019). This heterogeneity within the group needs to be ensured in the model, as was previously discussed in the methodology for SRQ2. In this example, a percentage of the agents of type 1 will still receive a TRUE as a result of the decision rules for the PCB component, enabling the behaviour. The decision rules thus remain the same for each agent, however the decisions change based on the environmental conditions.

The final model will spatially show agents which represent the number of tourists that partake in an activity of the group where that agent belongs to. It will therefore also be possible to show the total number of tourists partaking in each type of activity, although this number can only be interpreted relative to the other scenarios. Each configuration of parameters is to be simulated a sufficient number of times to account for the stochastic variation of the model (Balbi et al., 2013). The model results are then averaged to provide a final result for different scenarios.

5. Time schedule

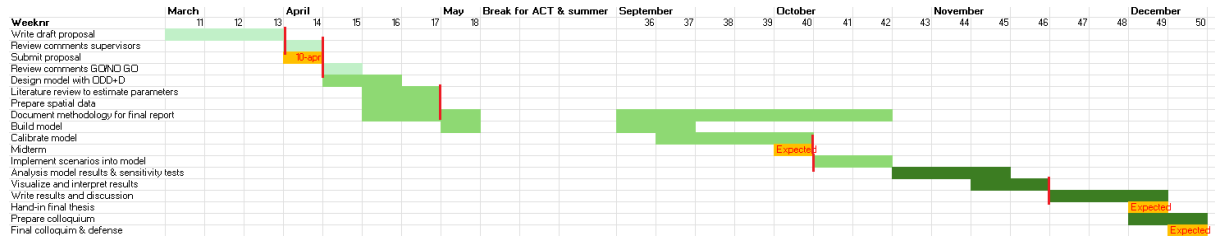


Figure 4 Gant chart planning with critical path

As can be seen from the Gant chart in Figure 4, I will work on my thesis in period 5 of academic year 2023-2024 and period 1&2 of academic year 2024-2025. To ensure a smooth start-up in period 1, I will make a final report for myself on all work done in period 5. The critical path is indicated with the red lines.

The expected midterm date is based on the first midterm date of this academic year, the expected colloquium data is also based on dates that were available this year.

6. Feasibility

6.1 Challenges

6.1.1 Calibration

The largest challenge lies in the model calibration, in which many factors must work together to create a system which functions like the real-world counterpart. Some factors which we want to model we do not know exactly, like how does the presence of a coffeeshop influence the tendency of an agent to visit? We can only make assumptions based on literature and observed patterns and see if these lead to the expected outcome.

To keep an understanding of the dynamics, the modelling will be done step by step and in an iterative way. First, the factors that seem to be most influential will be implemented into the model and the model will be ran. With additional elements, we want to come closer to the expected pattern. If additional elements that were thought influence tourist behaviour make the model show unrealistic outcomes, these can be removed. An unrealistic outcome is considered an agglomeration of tourists in places that are not generally known to have a lot of tourism, as much tourists in non-touristic areas as known touristic areas (like the Wallen) or a clear pattern of decreasing tourism under constant environmental conditions in the model.

It is also a possibility to reduce complexity of the model by reducing the heterogeneity of agents. In case the model turns out to be very sensitive to parameter with large uncertainty, this parameter will have to be removed.

6.1.2 Data availability for verification

Another challenge lies in that we do not know exactly how tourists move through Amsterdam. We only know the relative spread of tourists, as the neighbourhoods have been divided into quadrants for the report on tourism pressure throughout the city. To overcome this, the results of our model will not be taken as exact numbers. Instead, our results will show the spread of tourism relative to the previous situation or relative to the other parts of the city.

6.2 Materials

NetLogo: Open-source multi-agent modelling environment under the GPL License. Will be used to create and run the model.

GAMA: Open-source modelling and simulation platform for development of spatial ABMs. Alternative for NetLogo if its GIS extension does not provide sufficient spatial functionality.

Arcgis: GIS-software for mapping & analytics. Will be used to preprocess spatial data and may be used for visualization of the results.

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